

**Question:**

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

**Reasoning + Answer:**

< reasoning process here > CD 2 1 is a receptor on B cells that plays a key role in the recognition of complement proteins bound to pathogens , especially viruses . It acts as a co -re ceptor in the B cell receptor ( BC R ) complex , enhancing the response to antige ns that are coated with complement components ( like C 3 d ). This is particularly important for the recognition and activation of B cells by viruses such as Epstein -B arr virus ( EB V ), which can use CD 2 1 signaling to promote B cell activation and support infection .

EBV is known to bind to CD 2 1 on B cells , and the interaction between EB V and CD 2 1 contributes to the efficient infection and spread of the virus in B lymphocytes . When CD 2 1 is genetically down regulated or absent , B cells are less responsive to EB V , disrupting the early stages of viral entry or immune detection , and thereby reducing EB V's ability to establish infection .

In contrast , HIV primarily targets CD 4 + T cells and does not rely on CD 2 1 for entry or replication . HIV interacts with CD 4 and co -re ceptors like CX CR 4 or C CR 5 , not CD 2 1 . Human papill om avirus ( HPV ) primarily infects epith elial cells and does not target B cells significantly , and measles virus mainly infects respiratory epith elial cells , not B cells in the context of CD 2 1 signaling .

Therefore , the genetic modification that prevents expression of CD 2 1 would most likely interfere with Epstein -B arr virus path ogenesis due to its specific dependence on B cells and interaction with CD 2 1 .

< answer > A . Epstein -B arr virus ( EB V ) < /answer >

